

Practical-2

Practice lab Assignment

The screenshot displays the CodeTANTRA IDE interface. On the left, a panel titled "2.1.1. List operations" contains the assignment instructions. The main editor shows a Python file named "listOps.py" with the following code:

```
1 list=[]
2 while True:
3     print("1. Add")
4     print("2. Remove")
5     print("3. Display")
6     print("4. Quit")
7     choice =int(input("Enter choice: "))
8     if choice ==1:
9         num =int(input("Integer: "))
10        list.append(num)
11        print("List after adding:",list )
12    elif choice ==2:
13        if not list:
14            print("List is empty")
15        else:
```

Below the code editor, performance metrics are shown: Average time 0.091 s, Maximum time 0.136 s. Test results indicate "2 out of 2 shown test case(s) passed" and "1 out of 1 hidden test case(s) passed". A table compares expected and actual outputs for Test case 1:

Expected output	Actual output
1. Add	1. Add
2. Remove	2. Remove
3. Display	3. Display
4. Quit	4. Quit
Enter choice: 1	Enter choice: 1
Integer: 11	Integer: 11

At the bottom, a terminal window is visible with buttons for "Prev", "Reset", "Submit", and "Next".

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2.1.2. Dictionary Operations0/11

Write a Python program to perform insertion, update, deletion, and traversal operations on a dictionary. An initial dictionary containing 10 predefined records is already given in the program.

Operations to be Performed:

1. Insertion – Insert a new key-value pair into the dictionary using user input.
2. Update – Update the value of an existing key using user input.
3. Deletion – Delete a specified key from the dictionary using user input.
4. Traversal – Traverse the final dictionary and display all key-value pairs.

Input and Output Format:

1. Read an integer representing the key to be inserted and a string representing its value. Insert this new key-value pair into the dictionary and display the dictionary after insertion.
2. Read an integer representing the key to be updated and a string representing the new value. Update the value of the specified key (only if it exists) and display the dictionary after the update.
3. Read an integer representing the key to be deleted. Delete the specified key from the dictionary (only if it exists) and display the dictionary after deletion.
4. Finally, traverse the dictionary and display all key-value pairs.

The program should also display the original dictionary before performing any operations. Each output should be printed with appropriate messages.

Note:

- All operations must be performed using dictionary methods.
- Perform deletion only if possible; leave the dictionary unchanged.
- Refer to the visible test cases for better understanding and strictly match with the input/outputs.

Sample Test Cases

DictOper...

```
1 # Initial dictionary with 10 predefined records
2 student = {
3     1: "Amit",
4     2: "Riya",
5     3: "Kiran",
6     4: "Neha",
7     5: "Arjun",
8     6: "Pooja",
9     7: "Rahul",
10    8: "Sneha",
11    9: "Vikram",
12    10: "Anjali"
13 }
14 print("Original Dictionary:", student)
15 key=int(input())
```

Average time0.025 sMaximum time0.026 s2 out of 2 shown test case(s) passed2 out of 2 hidden test case(s) passed

Test case 128 msDebugExpected outputOriginal Dictionary: {1: 'Amit', 2: 'Riya', 3: 'Kiran', 4: 'Neha', 5: 'Arjun', 6: 'Pooja', 7: 'Rahul', 8: 'Sneha', 9: 'Vikram', 10: 'Anjali'}Actual outputOriginal Dictionary: {1: 'Amit', 2: 'Riya', 3: 'Kiran', 4: 'Neha', 5: 'Arjun', 6: 'Pooja', 7: 'Rahul', 8: 'Sneha', 9: 'Vikram', 10: 'Anjali'}After Insertion: {1: 'Amit', 2: 'Riya', 3: 'Kiran', 4: 'Neha', 5: 'Arjun', 6: 'Pooja', 7: 'Rahul', 8: 'Sneha', 9: 'Vikram', 10: 'Anjali', 11: 'Suresh'}After Insertion: {1: 'Amit', 2: 'Riya', 3: 'Kiran', 4: 'Neha', 5: 'Arjun', 6: 'Pooja', 7: 'Rahul', 8: 'Sneha', 9: 'Vikram', 10: 'Anjali', 11: 'Suresh'}

TerminalTest cases

PrevResetSubmitNext

Lab Assignment

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2.2.1. Linear search Technique0/23

Write a program to check whether the given element is present or not in the array of elements using linear search.

Input format:

- The first line of input contains the array of integers which are separated by space
- The last line of input contains the key element to be searched

Output format:

- If the element is found, print the index. If the element found multiple times, print the starting index
- If the element is not found, print **Not found**.

Sample Test Case:
Input:
1 2 3 4 3 5 6
3
Output:
2

Sample Test Cases

CTP1709...

```
1 arr = (input())
2 items = arr.split(" ")
3 s = (input())
4 if s in items:
5     print(items.index(s))
6 else:
7     print("Not found")
```

Average time0.011 sMaximum time0.014 s2 out of 2 shown test case(s) passed2 out of 2 hidden test case(s) passed

Test case 114 msDebugExpected output1 2 3 4 3 5 632Actual output1 2 3 4 3 5 632

Test case 212 ms

TerminalTest cases

PrevResetSubmitNext

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2.2.2. Captain of the Team

0.05 s

You are provided with the heights of 11 cricket players (in centimeters). Your task is to identify the tallest player, who will be selected as the captain of the team.

Input Format:
The first line of input will contain 11 integers, each representing the height of a player (in centimeters), each separated by a space.

Output Format
The output should be the height (in centimeters) of the tallest player.

Sample Test Cases

captainof...

1height=list(map(int,input().split()))

2tallest=max(height)

3print(tallest)

Average time0.008 s7.67 ms

Maximum time0.009 s9.00 ms

1 out of 1 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 10 msDebug

Expected output

171 169 185 156 174 191 186 190 187 172 160

191

Actual output

171 169 185 156 174 191 186 190 187 172 160

191

Terminal

Test cases

< Prev

Reset

Submit

Next >